Classification Model Evaluation Metrics

Overview

- Sensitivity and Specificity
- ROC and AUC
- Accuracy, Precision, Recall, F1-Score

Sensitivity and Specificity

SENSITIVITY AND SPECIFICITY

What are they?

How are they calculated?

TESTS → not perfect

Presence or absence of disease

How good is the test?

DISEASE

TEST	Disease	No disease
+	✓ True positive (TP)	✗ False positive (FP)
-	✗ False negative (FN)	✓ True negative (TN)

not always perfect

False positive → more testing
treatment
psychological effects
Cost
Risk

False negative ↓ delays in diagnosis and treatment
false sense of security
risky behaviour
legal consequences

DISEASE

TEST	Disease	No disease
+	90 True positive (TP)	100 False positive (FP)
−	10 False negative (FN)	400 True negative (TN)
600 people:	100	500

Type I Error

Type II Error

$$\text{SENSITIVITY} = \frac{TP}{TP+FN} = \frac{90}{100} = 0.9\ (90\%)$$

$$\text{SPECIFICITY} = \frac{TN}{FP+TN} = \frac{400}{500} = 0.8\ (80\%)$$

High sensitivity → screening tests (low false negatives)

High specificity → confirmatory tests (low false positives)

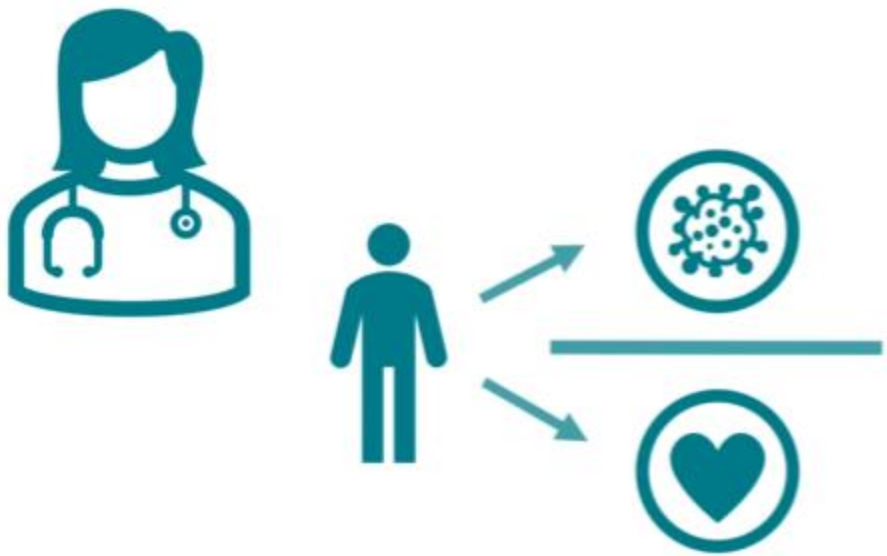
PERFECT TEST

SENSITIVITY 100%

SPECIFICITY 100%

ROC and AUC

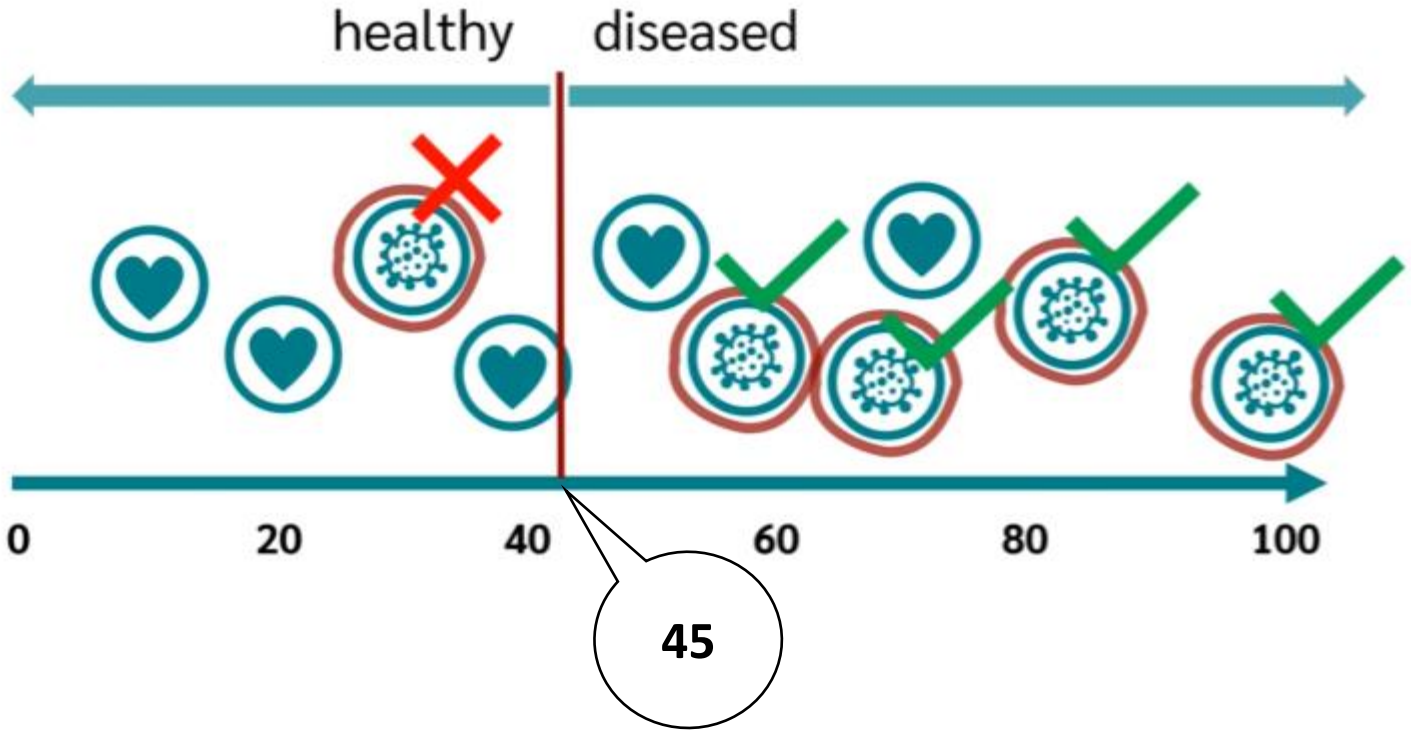
We would like to **classify** based on a **screening,** whether a person **has cancer** or **not.**



For this, we get **data from 10 people** about how high the **blood level** is and whether there is a **disease or not**.

This value is called the **True Positive Rate (TPR)** and is equal to the **sensitivity.**

So, we **correctly** classified **4 out of 5 as "diseased".**

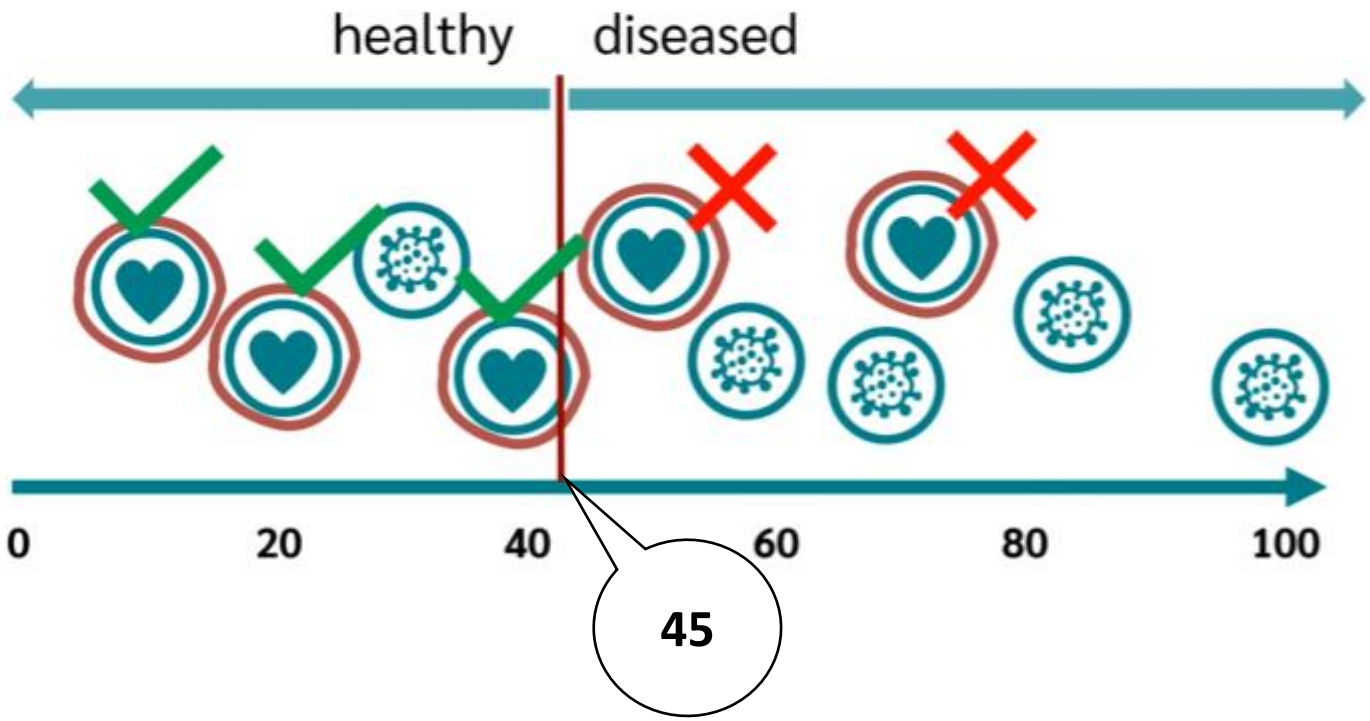


And it is precisely these **two values** that are then plotted on the **ROC Curve**.

So, for a **threshold of 45**, we get a **True Positive Rate** of **4/5** i.e. **0.8** and a **False Positive Rate** of **3/5** i.e. **0.6**.

So, we misclassified **2 out of 5 as "diseased“.** This value is called the **False Positive Rate (FPR)** and is 1 - specificity.

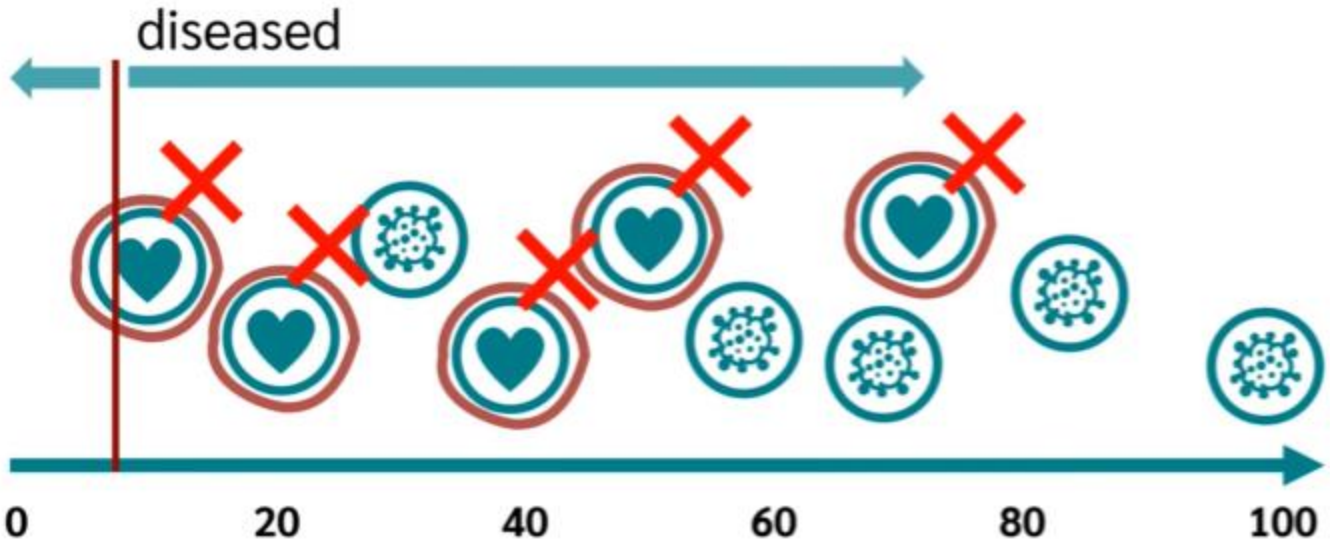
We can now calculate for **each threshold**, the **True Positive Rate** and the **False Positive Rate**.

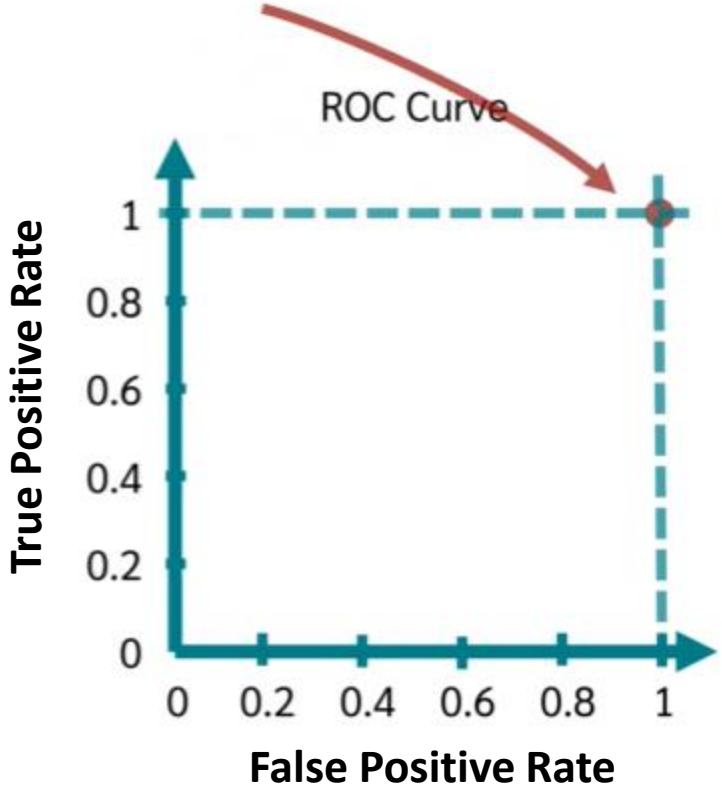


If we choose the **threshold value** to be very small,

i.e., pushed all the way to the **left**,

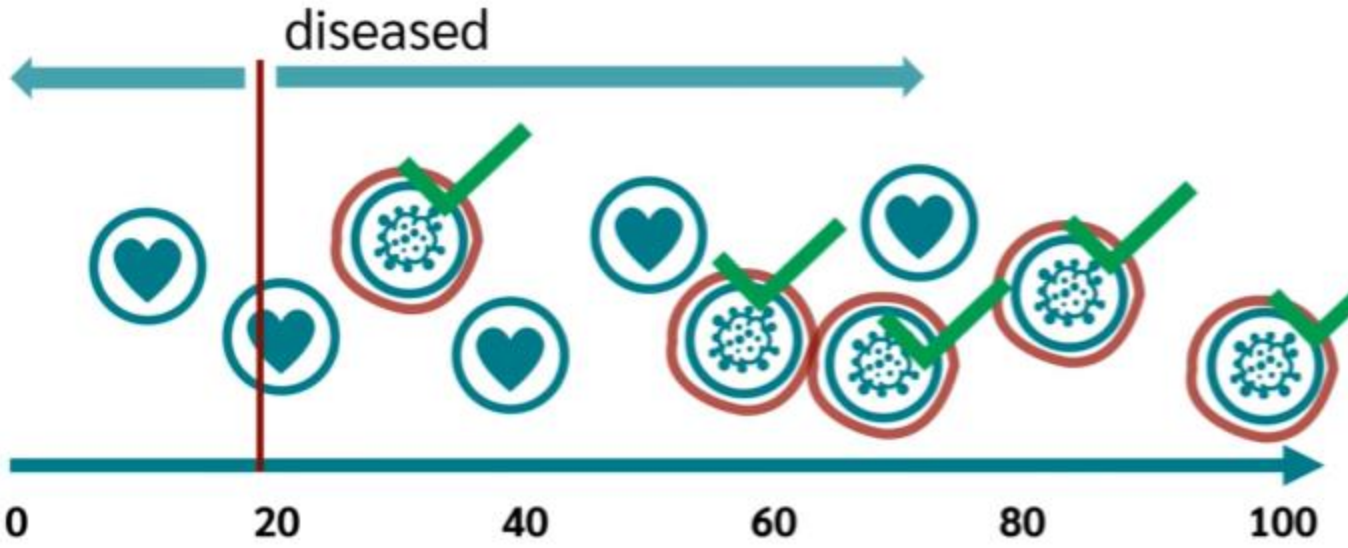
With that, we have the **first point**.

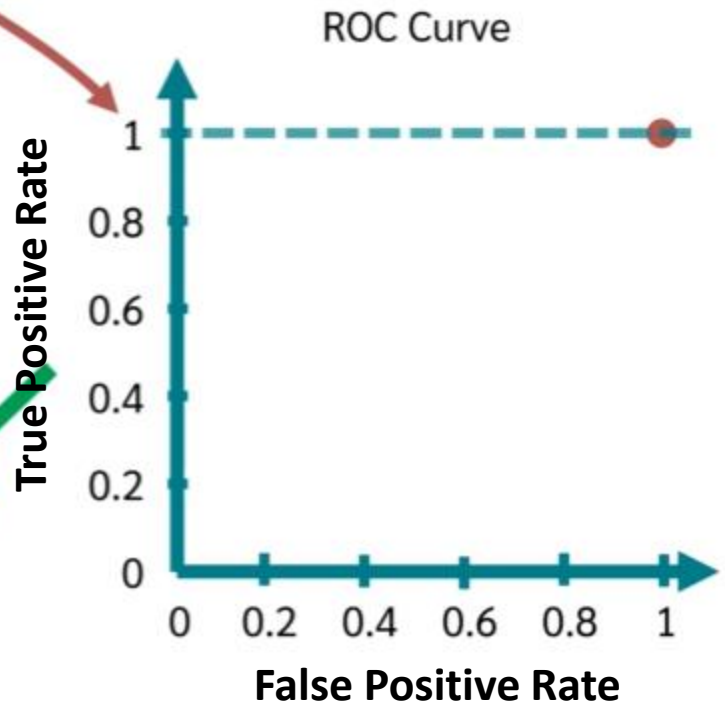




Here we still **correctly classify** all **5 diseased** people as **"diseased"**.

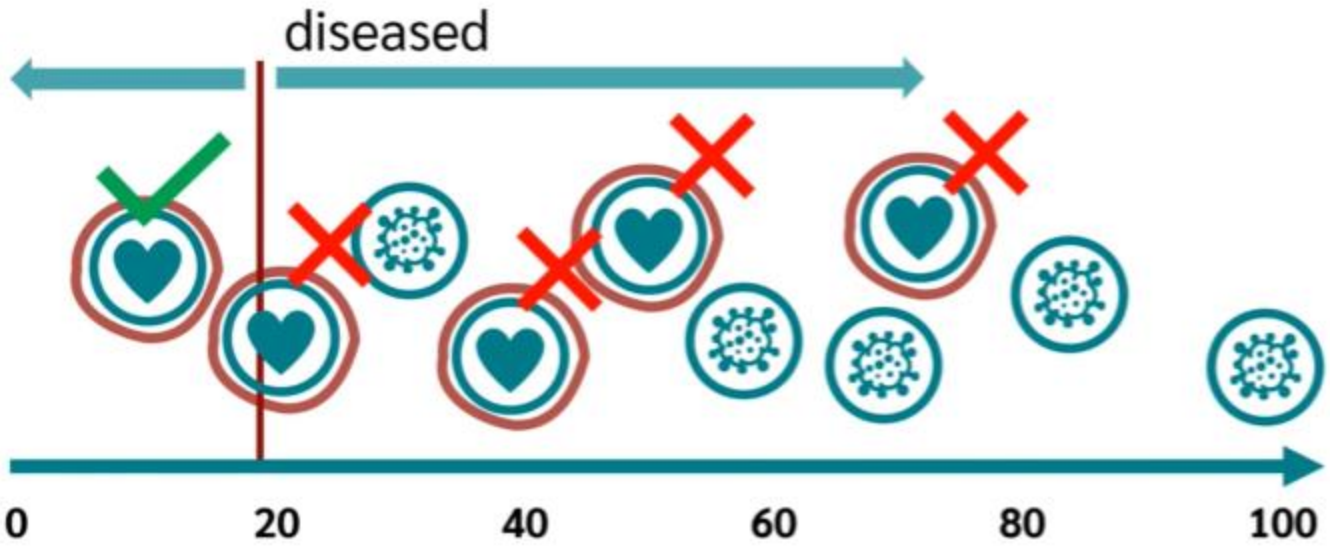
So, we continue to have a **True Positive Rate 5/5**.

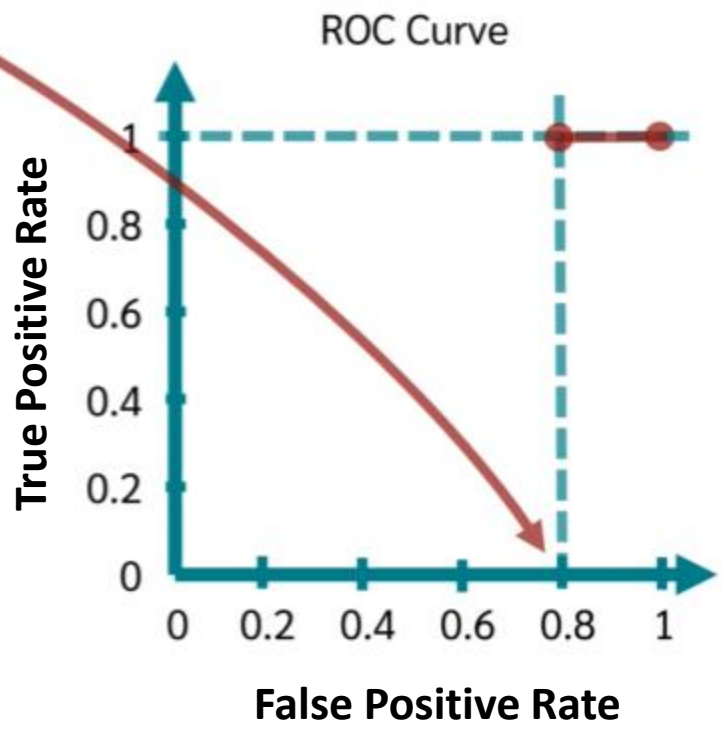




However, of the **5 healthy individuals**, we now **misclassify only 4 out of 5** as "diseased."

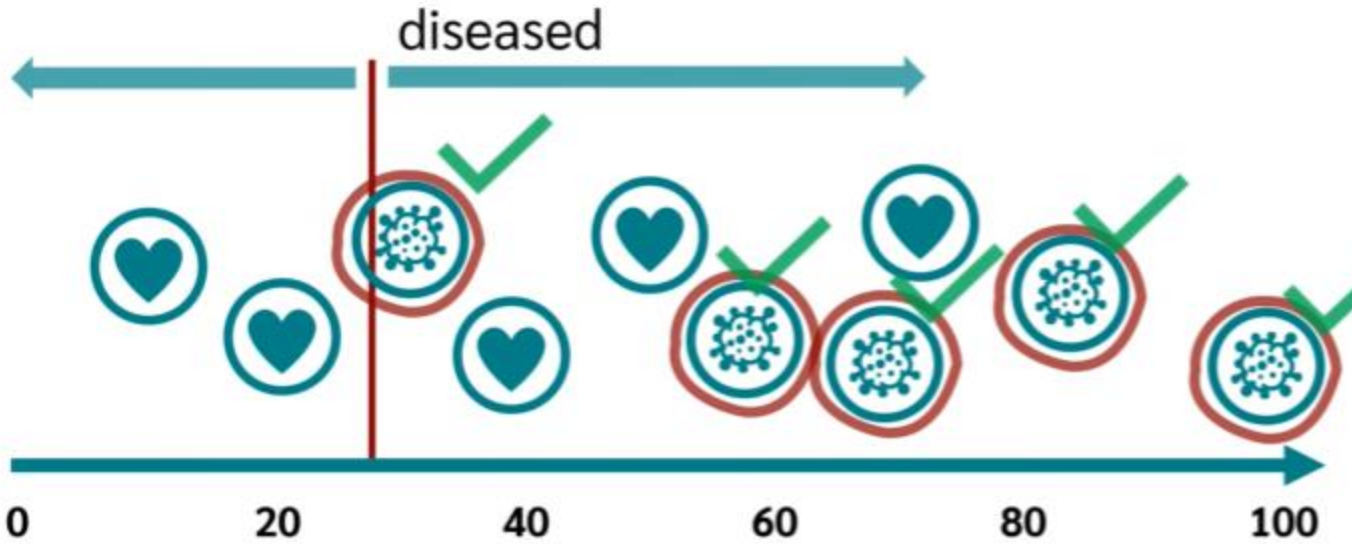
So, we have **4** out of **5** or **0.8**.

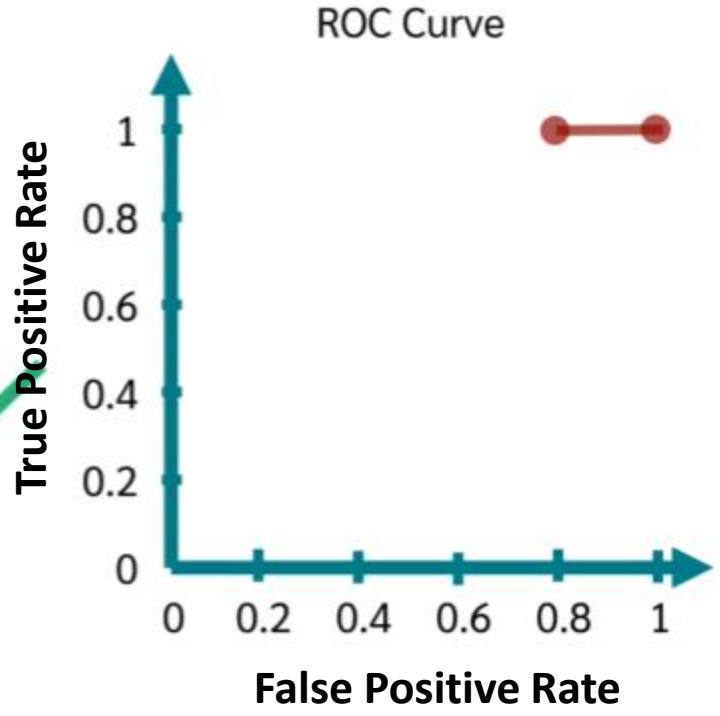




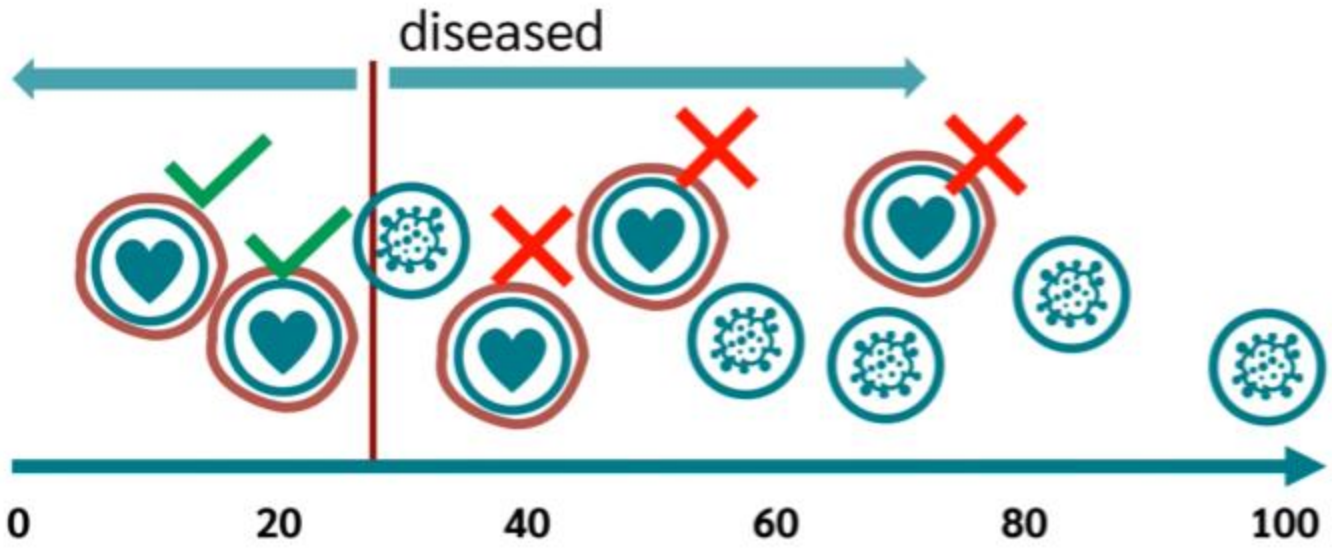
At the next **threshold,** we still have a **True Positive Rate of 1**.

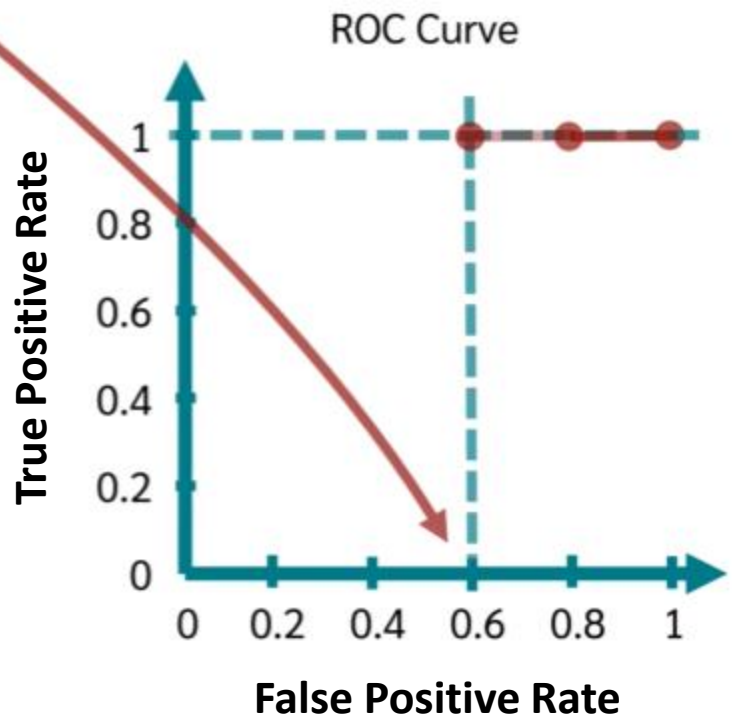
All **5 diseased** people are correctly classified.



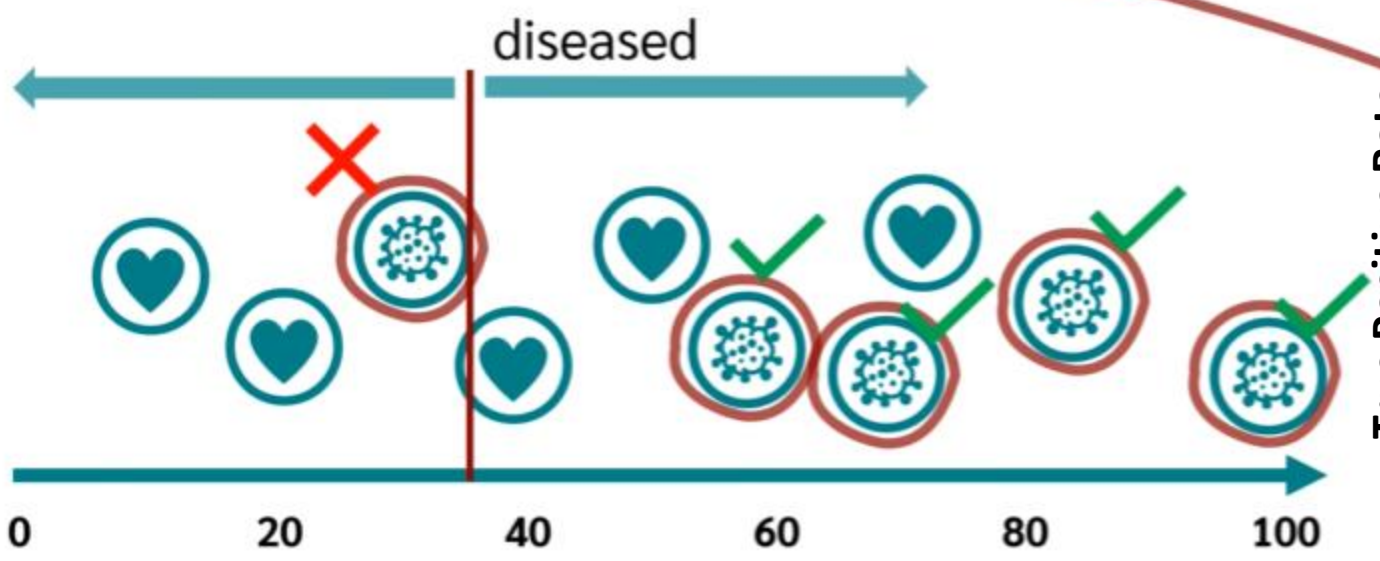


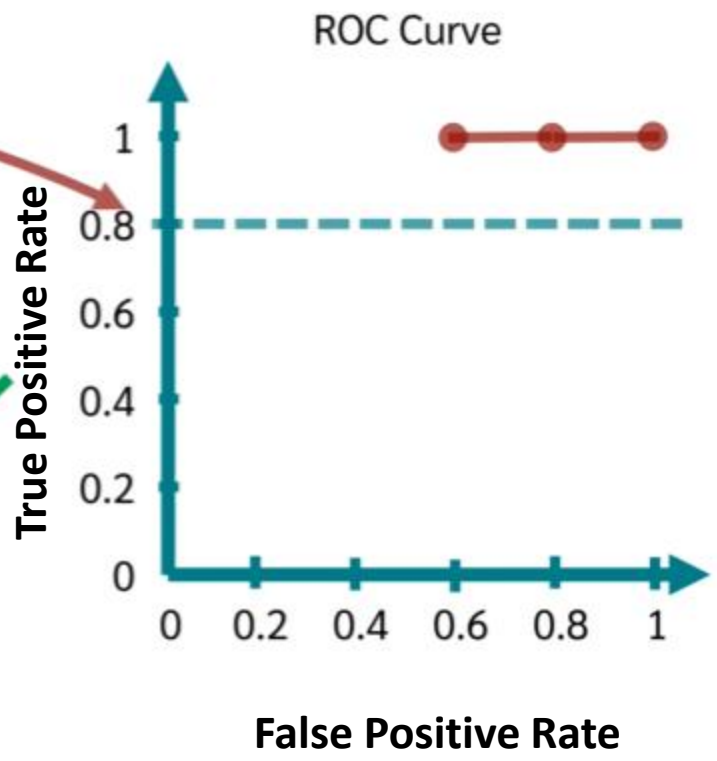
And a **False Positive Rate** of **3/5**, i.e. **0.6**





We therefore obtain a **True Positive Rate** of **4/5**, i.e. **0.8**.





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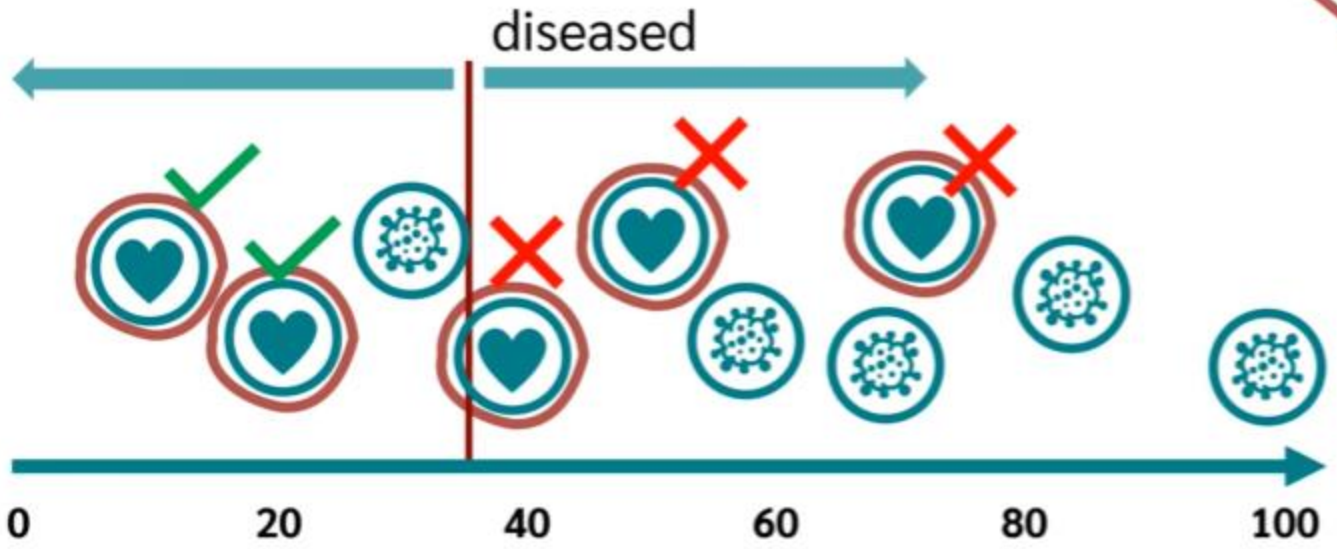
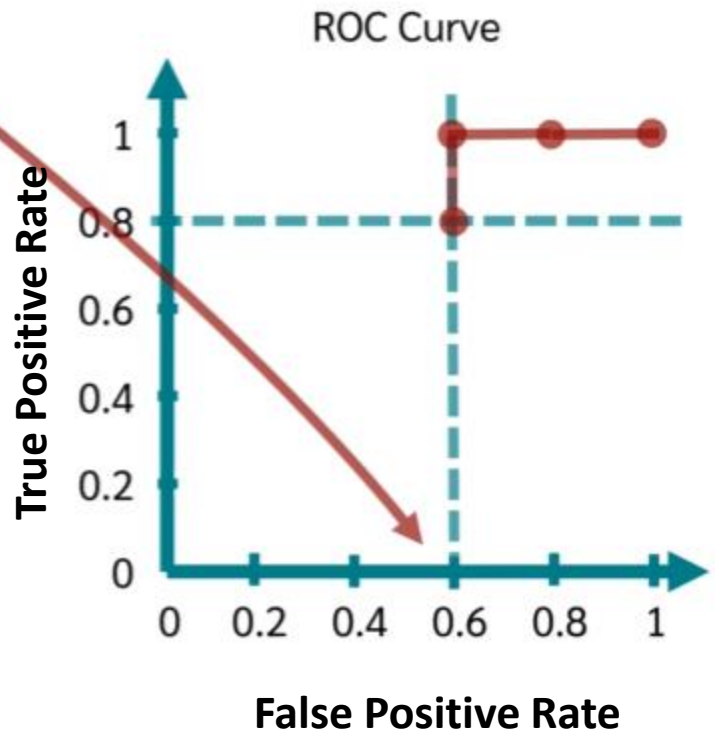
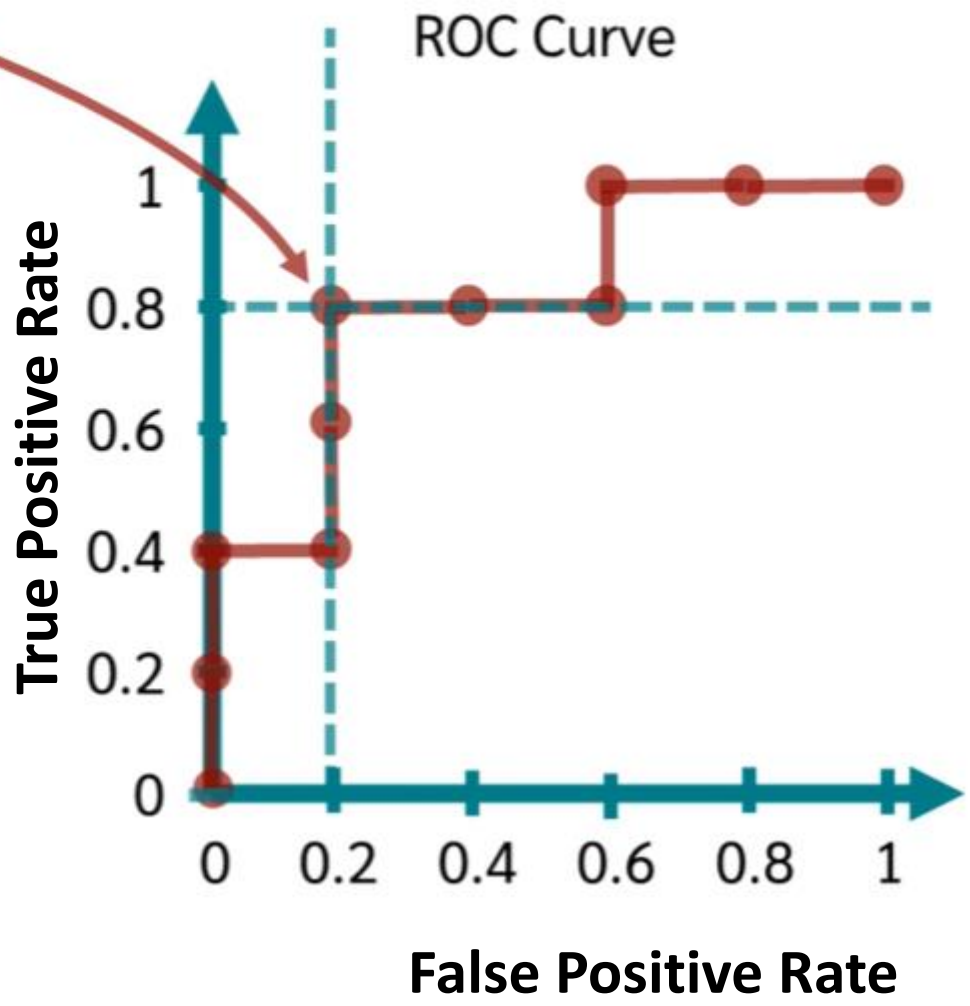




At this point, for example

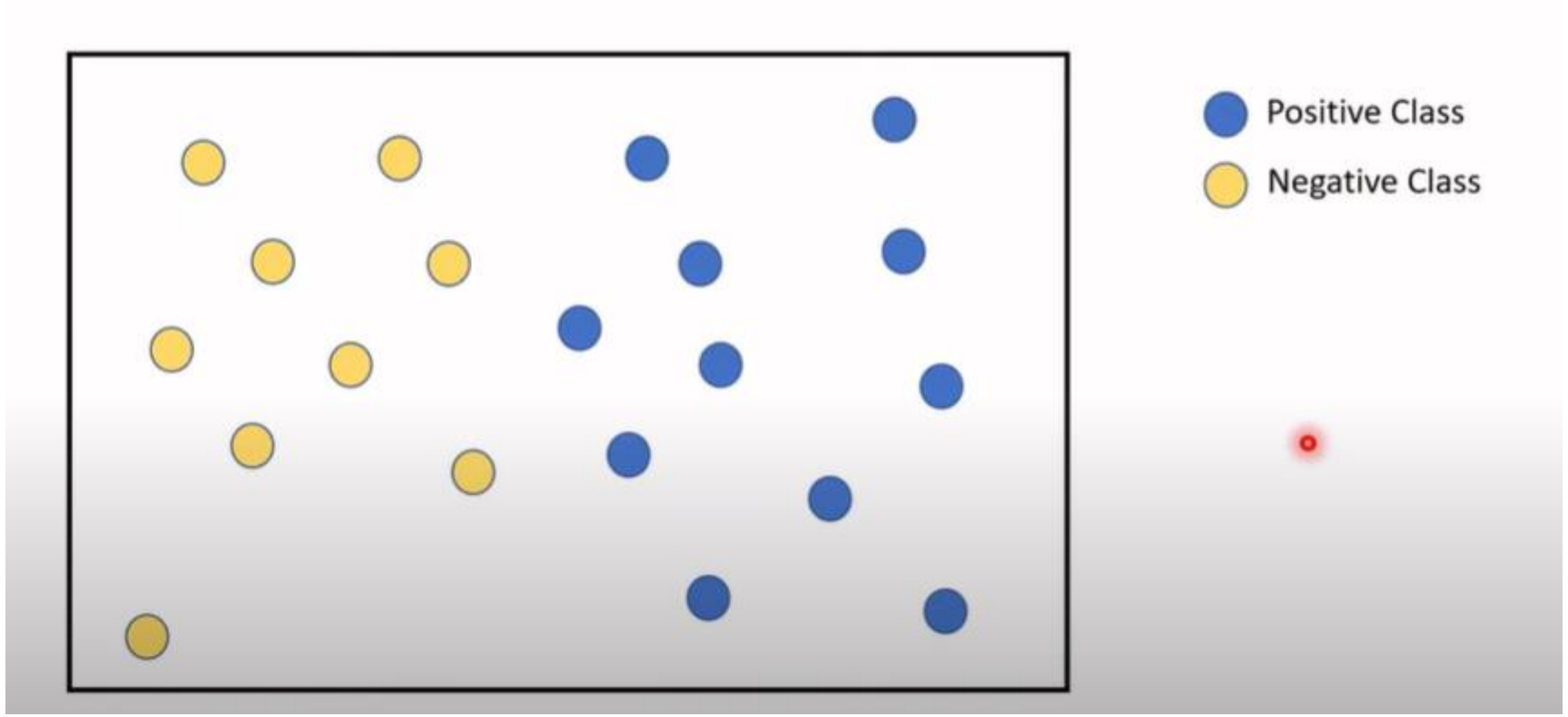
80% of the diseased individuals were correctly classified as “diseased“

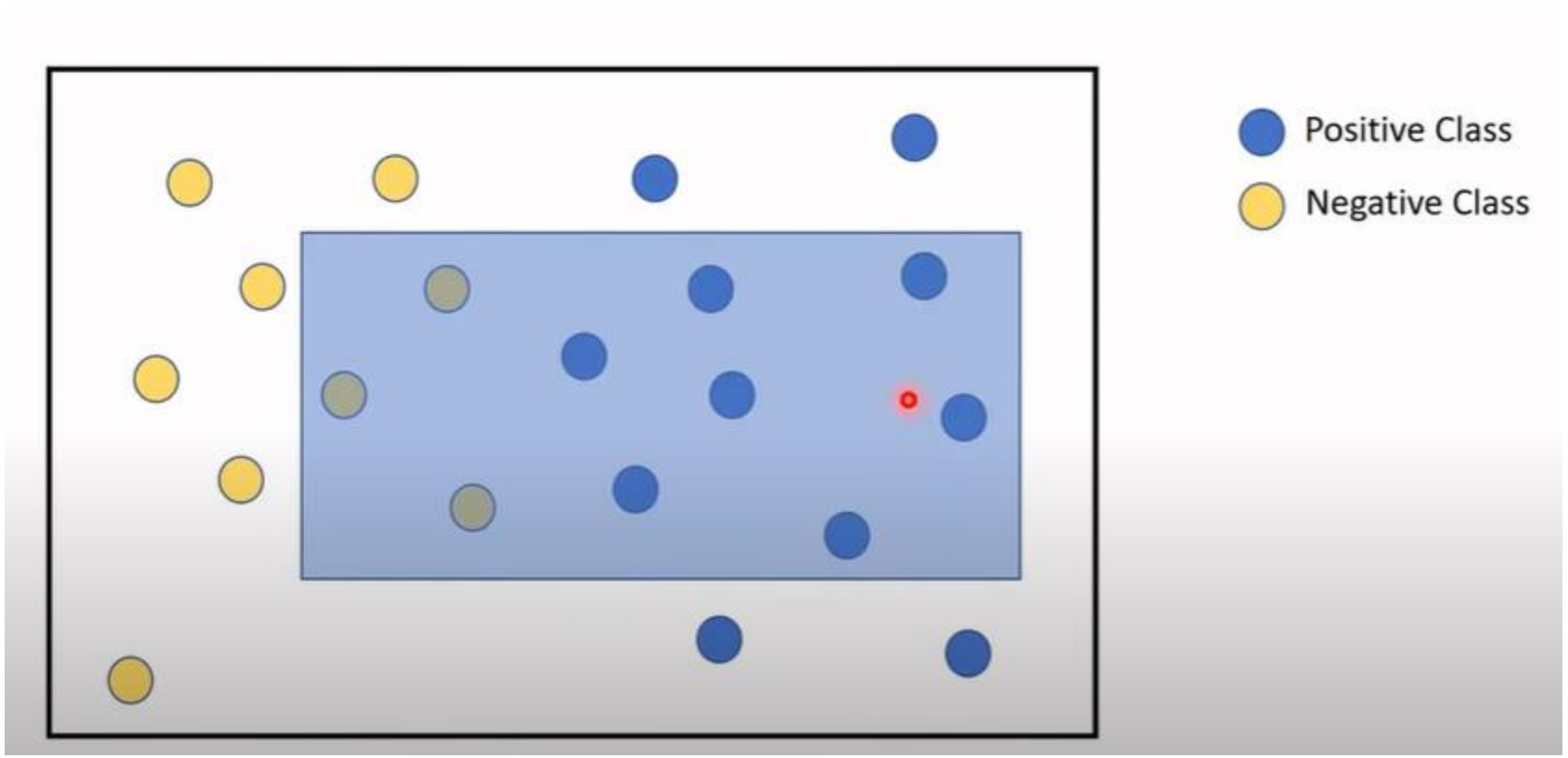
and **20%** of the healthy individuals were incorrectly assigned “diseased“.

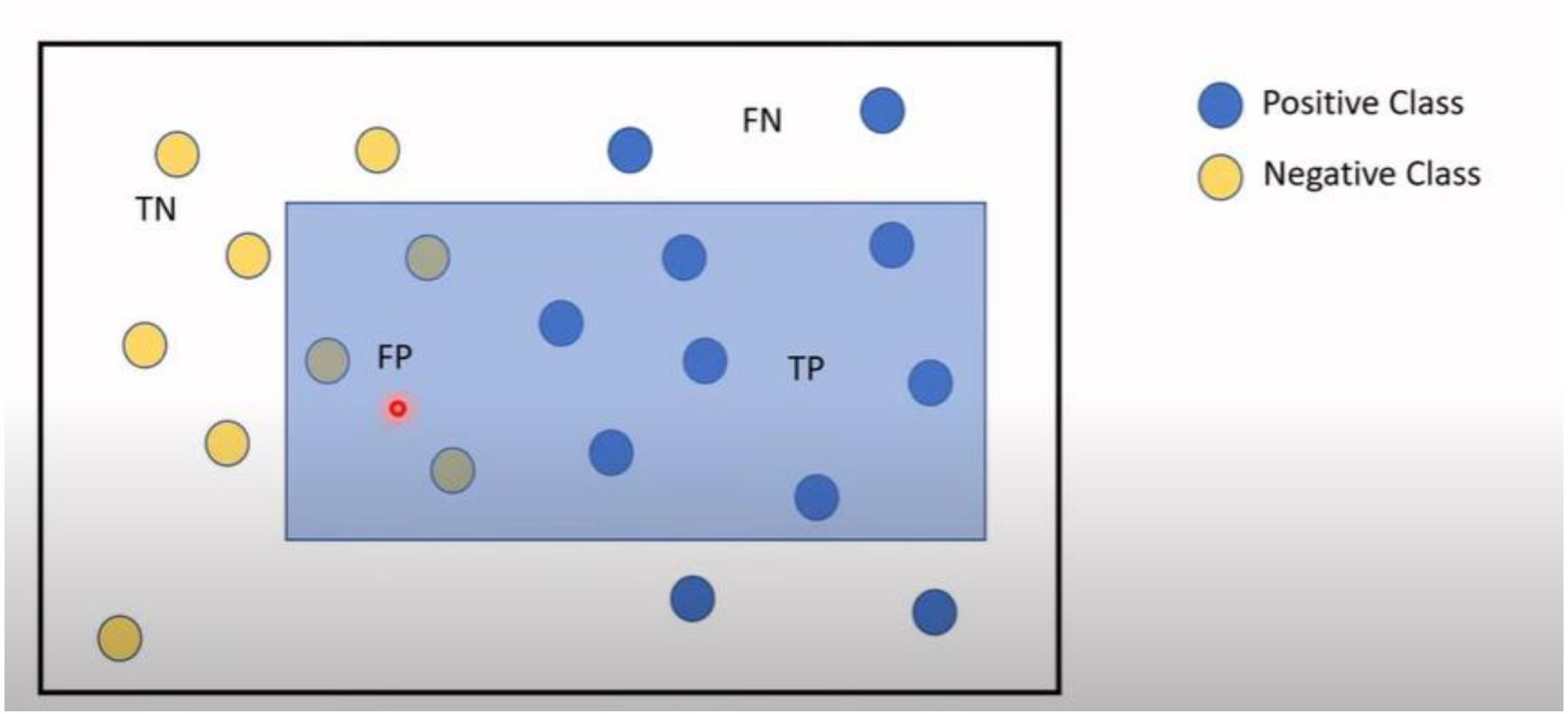


A **classification model** is better the **higher** the **curve**.

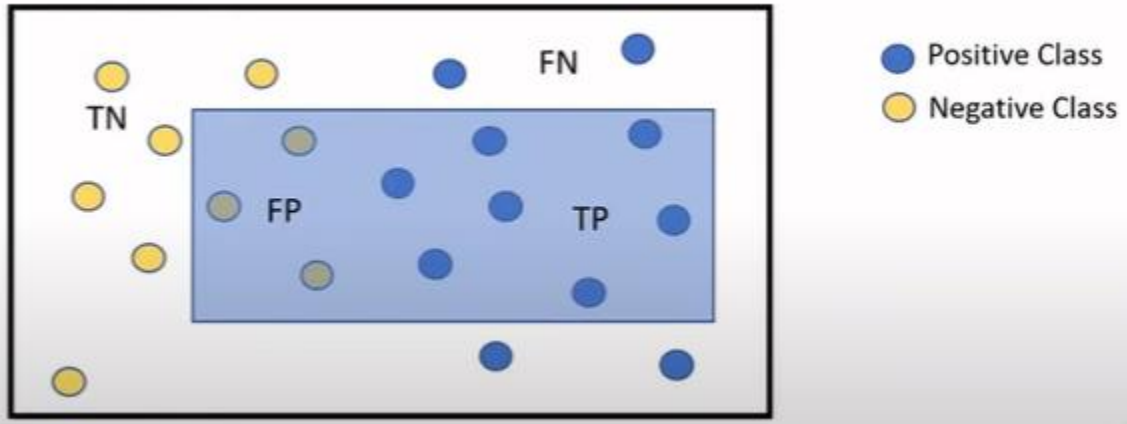
Accuracy, Precision, Recall, F1-Score


Positive Class
Negative Class

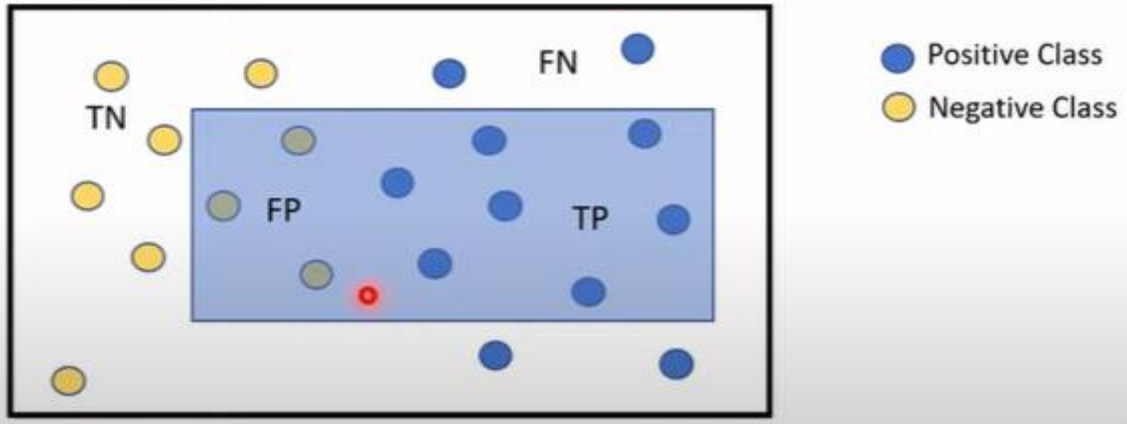

Positive Class
Negative Class


FN
TN
FP
TP
Positive Class
Negative Class

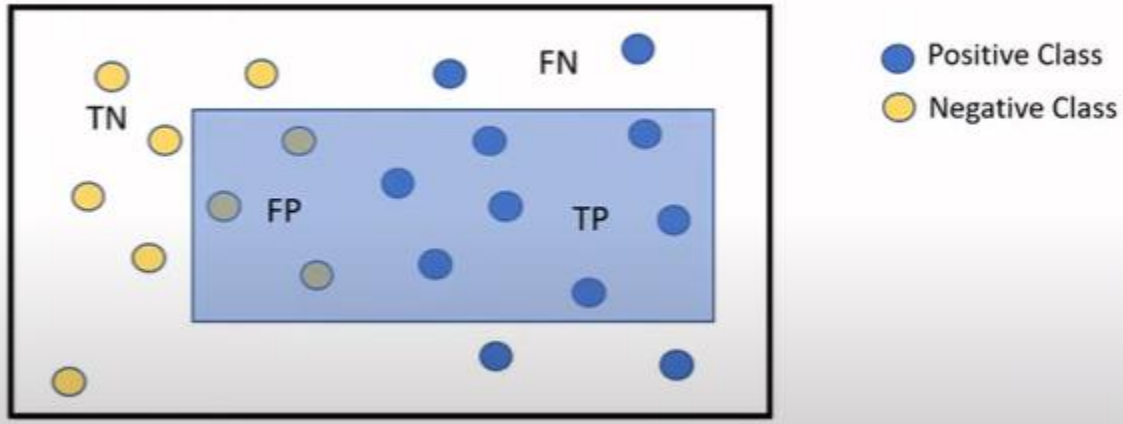
- $Accuracy = \frac{TP+TN}{TP+FP+FN+TN}$



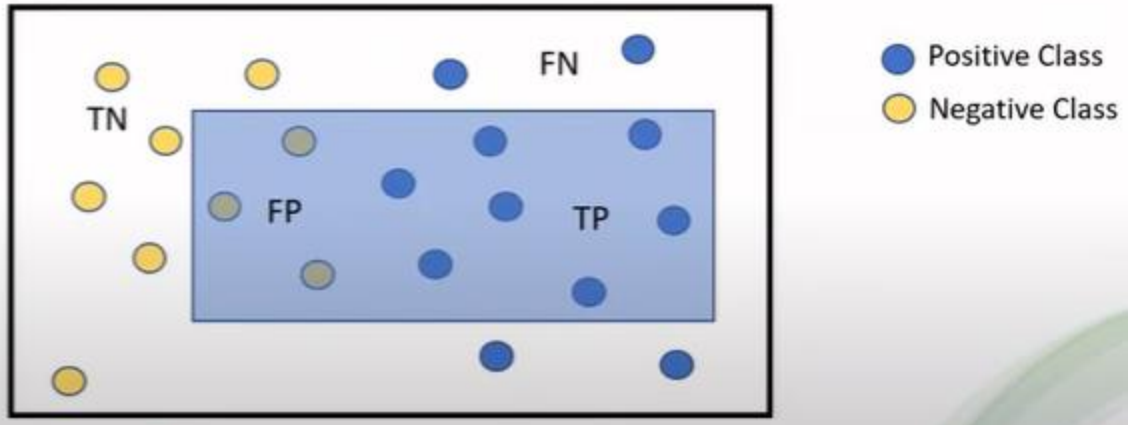
- $Precision = \frac{TP}{TP+FP}$



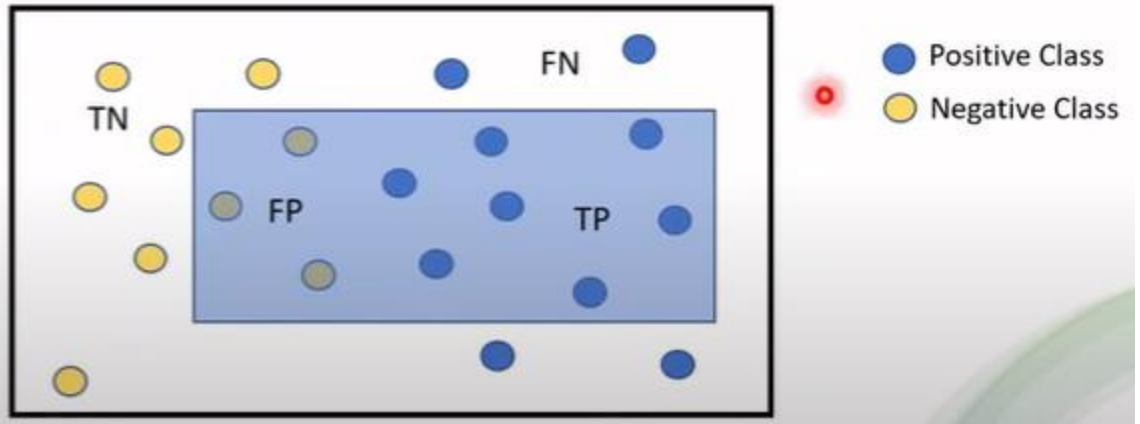
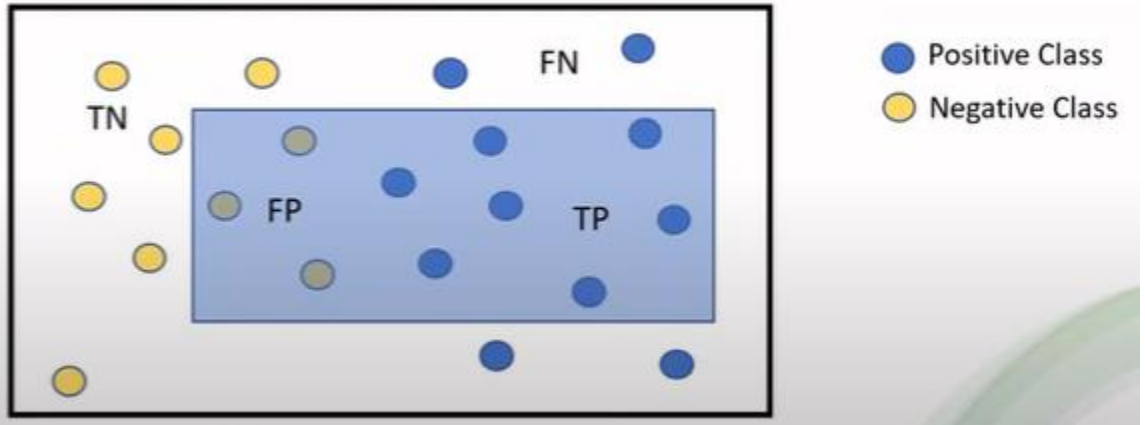
- $Recall = \frac{TP}{TP+FN}$



- $F1 - score = 2 * \frac{(Recall * Precision)}{(Recall + Precision)}$



- Sensitivity is also referred to as: true positive rate (TPR) and is the same as recall
- $Sensetivity = \frac{TP}{TP+FN}$



- Specificity is also referred to as: true negative rate (TNR)
- $Specicity = \frac{TN}{TN+FP}$